ATLAS: AND-Gated Task-Circuit and Loss-Aligned Saliency for Low-Bit LLM Quantization

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Abstract

Low-bit post-training quantization (PTQ) compresses large language models but often destroys task-specific capabilities when only a tiny fraction of weights can remain in full precision. Task-Circuit Quantization (TaCQ) selects FP16 outlier weights using contrastive gradients on cross-entropy loss, treating the model as a black box. We propose ATLAS (AND-gated Task-circuit Loss-Aligned Saliency): protect a weight only if it is salient both to the task loss and to the task circuit. Our first instantiation, ATLAS-T, attributes quantization sensitivity through sparse, approximately monosemantic transcoder features and intersects this circuit signal with standard CE saliency. Under a fixed 0.35% outlier budget matched to TaCQ on Llama-3.1-8B-Instruct, ATLAS-T improves Spider text-to-SQL execution accuracy by +2.9 points on average over four paired calibration seeds (29.4% vs. 26.5%) and GSM8K 2-bit accuracy from 25.4% to 28.7%. We then ask why it works. A controlled saliency-source ablation—everything matched, only the weight-ranking signal varies across eight arms—dissects the method and yields three findings: (i) task-circuit information dominates magnitude, quantization-error, and random placement (by 10–26 points at 2-bit); (ii) the conjunction of task loss and task circuit beats either signal alone; and (iii) the transcoder itself is not the active ingredient. Replacing it with the model's own MLP neurons as the feature basis (ATLAS-N) recovers most of the gain on GSM8K (26.5%) and surpasses all transcoder arms on MMLU-STEM (44.2% vs. 38.9%)—turning a method gated on external interpretability artifacts into one that applies to any transformer out-of-the-box. At 3-bit all task-conditioned methods converge, identifying conjunctive saliency as a distinctly low-bit phenomenon: where to spend the FP16 budget matters most exactly when the budget is harshest.

1 Introduction

Large language models (LLMs) are increasingly deployed under strict memory and latency budgets. Post-training quantization (PTQ) maps FP16 weights to 2–4 bits without retraining, but aggressive quantization—especially at 2 bits—can collapse performance on structured tasks such as text-to-SQL, math reasoning, and code generation. Mixed-precision PTQ mitigates this by keeping a small set of outlier weights in FP16 while quantizing the rest [Dettmers et al., 2022, Frantar et al., 2023, Dettmers et al., 2024].

The core problem. Outlier selection is a localization problem: which weights matter for a downstream task once quantization noise is injected? Uniform salience heuristics (magnitude, activation scale) ignore task structure. Calibration on generic corpora misses task-specific circuits. Even strong mixed-precision methods struggle in the 2-bit regime, where the allowed FP16 budget (often $<1\%$ of weights) must be spent with surgical precision.

Polysemantic weights vs. monosemantic features. A separate challenge is where in the network task signal lives. Individual MLP neurons and their connecting weights are typically polysemantic—each participates in many unrelated computations via superposition [Elhage et al., 2021, Bricken et al., 2023]. A gradient on cross-entropy therefore reflects a mixture of all roles a weight plays across the vocabulary and prompt positions, not a single task-specific mechanism. TaCQ inherits this ambiguity: although it conditions on task calibration data, saliency is still computed directly on physical weights from CE gradients, so the selected outliers may protect broadly useful directions rather than the sparse features that implement SQL grounding, aggregation, or schema linking.

Task-Circuit Quantization (TaCQ). Xiao et al. [2025] frame outlier selection as automated circuit discovery for PTQ. They score each weight by combining (i) contrastive gradients on task calibration data, (ii) expected weight change under quantization $\Delta W = W - W_{\text{quant}}$, and (iii) a global top-p mask under a fixed memory budget. Crucially, TaCQ operates in weight space on polysemantic gradient signal; it does not decompose MLP computation into features before ranking outliers.

Step 1 — ATLAS-T: route saliency through a monosemantic basis. Our starting point is interpretability-guided: if CE gradients on weights are polysemantic, attribute through a basis that is not. We call the resulting family ATLAS (AND-gated Task-circuit Loss-Aligned Saliency); its first instantiation, ATLAS-T (ATLAS with a Transcoder basis), uses sparse TopK transcoders [Dunefsky et al., 2024] to identify features that activate preferentially on task-faithful versus corrupted inputs, back-propagates a feature-filtered reconstruction objective, and intersects the resulting circuit saliency with standard CE saliency (geometric mean; §5). This works: +2.9 points mean Spider execution accuracy over four paired seeds and +3.3 points GSM8K at 2-bit versus TaCQ at matched budget, with the selected mask sharing only J=0.30 of its weights with TaCQ's.

Step 2 — dissect it: which ingredient carries the gain? A method with three coupled ingredients (contrastive circuit localization, a pretrained transcoder dictionary, and a conjunction rule) invites the question of what is actually doing the work—and transcoders exist only for a handful of models, which would cap the method's reach. We therefore run a saliency-source ablation in which everything is held fixed (model, 0.35% budget, contrastive pairs, GPTQ, evaluation) and only the signal that ranks weights varies across eight arms: random, weight magnitude $|W|$, magnitude $\times$ quantization error $|W| \cdot |\Delta W|$, CE ($\approx$TaCQ), circuit-only, and the conjunction—each circuit arm instantiated in two bases: the transcoder and, replacing it, the model's own MLP neurons with down_proj as the decoder.

Step 3 — the distilled method: ATLAS-N. The ablation delivers a clean dissection: the conjunction is the active ingredient, and the dictionary is not. The native-basis conjunction, ATLAS-N (ATLAS with the model's Native MLP-neuron basis), recovers most of ATLAS-T's gain on GSM8K and surpasses it on MMLU-STEM, while requiring no external dictionary, SAE, or transcoder—the method applies out-of-the-box to any transformer. What began as a transcoder-dependent technique ends as a general principle: protect a weight only if it is salient to both the task loss and the task circuit; any sufficiently task-discriminative feature basis—including the model's own—suffices to define the circuit.

Contributions.

- ATLAS-T: we introduce transcoder-guided outlier selection for PTQ—routing saliency through task-discriminative monosemantic features and intersecting with CE—and show it beats TaCQ at matched budget on Spider (+2.9 pp mean over four paired seeds) and GSM8K 2-bit (+3.3 pp), with interpretability analysis showing low mask overlap (J=0.30) and causal knockout (−29 pt).
- Dissection: a controlled eight-arm saliency-source ablation isolates the active ingredient. Task-circuit information dominates magnitude/error/random placement by 10–26 points at 2-bit, and the conjunction beats both of its parts (GSM8K: 28.7% vs. CE 25.4% / circuit-only 25.6%; §7.3).
- ATLAS-N: the dissection shows the dictionary is optional. Using the model's own MLP neurons as the circuit basis recovers most of the transcoder gain on GSM8K and surpasses it on MMLU-STEM (44.2% vs. 38.9%), removing the dependency on external interpretability artifacts and extending task-circuit PTQ to any transformer.
- Regime analysis: all task-conditioned methods converge at 3-bit; conjunctive saliency is a low-bit phenomenon (§7.4), locating exactly where saliency choice matters.
- We release code, contrastive calibration sets, and reproduction scripts for TaCQ, ATLAS-T, and ATLAS-N masks. [TODO: anonymize for submission]

Paper roadmap. Section 2 gives the theoretical framing; Section 3 states research questions; Section 4 situates our work; Section 5 describes the algorithm and the dictionary-free variant; Sections 6–7 cover setup and findings.

2 Theoretical Framing: Encoded–Used Gap Dissociation

We frame ATLAS-T against TaCQ as a targeted scalpel vs. a global broadsword. Both preserve a tiny FP16 outlier budget under mixed-precision GPTQ; they differ in what signal selects those outliers.

The encoded–used ambiguity in TaCQ. Cross-entropy on final logits conflates two roles of weights in text-to-SQL:

- Encoding: weights that store schema syntax, SQL templates, and lexical priors (often active whenever SQL-like tokens appear).
- Using: weights on causal execution edges—routing that binds columns, applies aggregations, and composes structure for the current question.

Because CE gradients flow through the full vocabulary, TaCQ's contrastive saliency still mixes encoding and use under a single polysemantic ranking [Xiao et al., 2025]. At 0.35% outliers, wasting budget on passive encoders directly costs execution accuracy on Spider.

Dissociation via F_{task}. ATLAS-T selects transcoder features with $\Delta a = \text{ReLU}(a_{\text{clean}} - a_{\text{corr}}) > \tau$ and builds y^{target} only from this task set F_{task}. Gradients therefore align with weights that implement contrastive, task-active MLP directions, not every pathway that lowers CE on SQL surface form. We call this encoded–used gap dissociation: outliers target active routing momentum rather than undifferentiated static memory. [TODO: Cite feature-level audit or layer analysis supporting dissociation.]

What we claim in this paper vs. systems extensions. This paper: same static mask, same quantizer, same budget as TaCQ; test whether dissociation improves Spider execution accuracy at 2–3 bits. Extensions (outlook, §11): distinct sub-task masks, orthogonality constraints, and KV-cache-preserving hot-swaps—architectural capabilities TaCQ's static monolith cannot express.

3 Research Questions

RQ1: Conjunctive vs. single-signal saliency. Does requiring a weight to be salient to both the task loss and the task circuit identify a better outlier set than either signal alone at the same FP16 budget? Status: Yes at 2-bit. Ablation (Table 5): mult 28.7% vs. ce 25.4% / align 25.6% on GSM8K; $\text{mult}_{\text{free}}$ 44.2% vs. ce 41.3% / $\text{align}_{\text{free}}$ 42.6% on MMLU-STEM. Spider: ATLAS-T +2.9 pp mean over four paired seeds (Table 2).

RQ2: Is an external dictionary necessary? Can the circuit term be computed in the model's native MLP-neuron basis without transcoders or SAEs? Status: Yes. ATLAS-N recovers 92% of the transcoder gain on GSM8K and surpasses all transcoder arms on MMLU-STEM (§7.3). The dictionary is an optional booster, not a dependency.

RQ3: Circuit vs. bit-placement heuristics. Do magnitude, quantization-error, and random selection explain the gains? Status: No. Controls trail circuit-informed arms by 10–26 points at 2-bit on GSM8K and sit near chance on MMLU-STEM (Table 5).

RQ4: Bit-width dependence. At what quantization severity does saliency choice matter? Status: Conjunctive saliency is a low-bit phenomenon: decisive at 2-bit, within seed noise at 3-bit (§7.4).

RQ5: Feature–weight alignment under superposition. Do task-discriminative circuit features map to distinct weight subsets from CE saliency? Status: J=0.298 (2-bit) / 0.331 (3-bit) between ATLAS-T and TaCQ; mid layers 4–14 and MLP projections diverge most; cross-seed stability J=0.995–0.997 (ATLAS-T) vs. 0.983–0.985 (TaCQ) (§9.1).

RQ6: Calibration source. How much does the calibration distribution matter at 2–3 bit? Status: A 25+ point collapse when switching TaCQ MSG from Spider to C4 calibration at 2-bit (§8); task-matched calibration is necessary even with the mask held fixed.

RQ7: Baseline reproduction. Does our ported TaCQ pipeline behave consistently with published numbers? Status: Llama-3-8B Spider replication matches paper targets (67.7 / 22.3 / 60.1 exec%).

4 Related Work

Post-training quantization. GPTQ [Frantar et al., 2023] remains the workhorse for weight-only PTQ of LLMs. Activation-aware methods (AWQ [Lin et al., 2024]), omnidirectional calibration (OmniQuant [Shao et al., 2024]), and 2-bit specialized schemes (QuIP# [Chee et al., 2024]) improve uniform quantization but still degrade on hard reasoning tasks at 2 bits. Mixed-precision and sparse formats—LLM.int8() [Dettmers et al., 2022], SPQR [Dettmers et al., 2024], SliM-LLM [Huang et al., 2025]—retain high-magnitude or salient weights in higher precision via magnitude, Hessian, or group-wise bit allocation. ReALLM [Bedin et al., 2025] compresses each weight matrix with a residual autoencoder plus vector quantization—a different architectural decomposition, not task-conditioned outlier selection. These methods generally define salience without an explicit task-circuit or monosemantic feature basis.

Task-conditioned mixed precision. TaCQ [Xiao et al., 2025] is the closest prior work to ours. Presented at COLM 2025, it uses contrastive calibration, quantization-aware localization (QAL) and magnitude-sharpened gradients (MSG) with a global top-p mask, explicitly connecting PTQ to automated circuit discovery and knowledge localization [Conmy et al., 2023]. Across QA, math, and text-to-SQL on Llama-3 and Qwen2.5, TaCQ outperforms SPQR and SliM-LLM at matched bit budgets—especially at 2–3 bits. We treat TaCQ as the primary baseline because it shares our GPTQ stack, ΔW corrupt model, outlier fraction (0.35%), and Spider evaluation protocol. However, TaCQ scores polysemantic weights via CE on logits; ATLAS-T intersects CE with saliency routed through a monosemantic transcoder basis (§5). Empirically, our ATLAS-T mask overlaps only 30% (Jaccard J=0.298) with TaCQ at the same budget (§??), showing the gain is not a minor reranking within TaCQ's weight pool.

Superposition, polysemanticity, and sparse features. Elhage et al. [2021] argue that capacity constraints force LLMs to encode more features than neurons, yielding polysemantic units whose activations mix many concepts. Dictionary learning and sparse autoencoders (SAEs) recover latents closer to monosemanticity—each feature aligns with a narrower behavior [Bricken et al., 2023, Cunningham et al., 2024]. Transcoders apply the same idea at MLP boundaries: they approximate MLP(x) with a wider, sparsely activating layer [Dunefsky et al., 2024], enabling cleaner circuit factorization than residual-stream SAEs [Wright et al., 2025]. Prior work uses these tools for analysis, steering, or verification (§4); we ask whether monosemantic feature attribution should drive compression by deciding which polysemantic weights must remain in FP16.

Mechanistic interpretability, transcoders, and circuit verification. Transformer circuits [Elhage et al., 2021] and automated circuit discovery [Conmy et al., 2023] seek sparse subgraphs explaining behaviors. Meta's Circuit Tracer and the public crv-8b-instruct-transcoders weights provide per-layer TopK transcoders for Llama-3.1-8B-Instruct [Ameisen et al., 2026]. Circuit-based Reasoning Verification (CRV) [Ameisen et al., 2026] replaces MLPs with these transcoders to build attribution graphs, classify reasoning errors, and intervene on individual features—the same sparse basis we use, but for runtime verification rather than offline weight selection. Unlike TaCQ, which ranks polysemantic weights from CE gradients, ATLAS-T attributes through this monosemantic feature basis first, then maps saliency back to physical parameters.

Table 1: Positioning vs. closest prior work (all target low-bit LLM compression or control).

Method	Goal	Feature basis	Weights?
SPQR / AWQ / SliM-LLM	PTQ	Magnitude / Hessian / salience	✓
TaCQ [Xiao et al., 2025]	Task PTQ	CE (polysemantic)	✓
SAS / SAE-TS [Bayat et al., 2025, Chalnev et al., 2024]	Steer	SAE latents	—
CRV [Ameisen et al., 2026]	Verify CoT	Transcoder latents	—
ATLAS-T (ours)	Task PTQ	Transcoder ∩ CE	✓

Sparse-feature steering and contrastive attribution. Contrastive prompts (clean vs. corrupted inputs) appear in influence estimation, data attribution, and TaCQ's gradient capture. Sparse Activation Steering (SAS) [Bayat et al., 2025], also at COLM 2025, uses contrastive prompt pairs to isolate behavior-specific SAE latents and steer model activations at inference— sharing our motivation to escape superposition, but modifying activations at test time rather than selecting weights at compression time. SAE-Targeted Steering [Chalnev et al., 2024] constructs steering vectors by modeling effects on SAE features; again, the output is a steering direction, not a quantization mask. We reuse contrastive calibration infrastructure but change what is being attributed: transcoder feature activations on MLP hooks (and their intersection with CE), combined with $|W| \cdot |\Delta W|$ as in TaCQ.

Quantization vs. interpretability. Recent work studies how PTQ damages SAE features even when perplexity looks stable [Anonymous, 2026], motivating feature-level audits of compressed models. Our direction is complementary: we use interpretability tools upstream to choose a mask that preserves task performance, then measure whether the selected circuit differs from CE-only selection.

Text-to-SQL as a testbed. Spider [Yu et al., 2018] is a standard semantic parsing benchmark requiring schema grounding and compositional SQL. It stress-tests structured output under quantization and matches TaCQ's headline results. We extend evaluation to [TODO: GSM8K, HumanEval] for generality.

Positioning. Table 1 summarizes how ATLAS-T relates to the closest lines of work. To our knowledge, no prior conference paper uses transcoder (or SAE) task-feature attribution to select PTQ outlier weights under a fixed FP16 budget and reports both (i) downstream task gains vs. TaCQ and (ii) low mask overlap showing a distinct circuit. ATLAS-T sits at the intersection of task-aware PTQ [Xiao et al., 2025, Huang et al., 2025] and monosemantic feature attribution [Dunefsky et al., 2024, Bayat et al., 2025, Ameisen et al., 2026]. Unlike TaCQ, we do not treat weights as atomic units of task importance—we decompose MLP computation to mitigate polysemanticity before selecting outliers. Unlike steering and verification work, we do not seek human-readable explanations or runtime control for their own sake—we convert feature-level task signal into a weight mask with measurable downstream impact (+3.4 pt mean Spider exec at 2-bit vs. TaCQ, four seeds; §??).

5 Method: ATLAS — AND-Gated Task-Circuit and Loss-Aligned Saliency

We keep the TaCQ mixed-precision quantization pipeline (GPTQ with per-weight bit-width mask) and replace only the saliency extraction stage.

Setup. Model f_θ with weights θ; uniform b-bit reference $W_{b\text{bit}}$ (RTN or GPTQ on task calibration data); outlier budget k (fraction of parameters kept in FP16, 0.35% throughout). Contrastive pairs (x^+, x^-) where x^+ is task-faithful and x^- is corrupted (schema errors, wrong CoT, wrong answer choice, etc.).

TaCQ saliency (review). Accumulate $|\nabla_W \mathcal{L}_{\mathrm{CE}}(x^+)|$ over calibration, multiply by $|W| \cdot |W - W_{b\mathrm{bit}}|$, global top-$k$ selection [Xiao et al., 2025]. Gradients are taken w.r.t. polysemantic weight tensors: each W may contribute to many superposed features, and $\mathcal{L}_{\mathrm{CE}}$ mixes all roles active at each token.

5.1 Circuit saliency in a feature basis

The circuit term inserts a feature-space bottleneck before weight saliency. For each layer ℓ, an encoder maps the MLP computation to a code $f \in \mathbb{R}^{F_\ell}$; a decoder D_ℓ maps codes back to the MLP output space. We rank task-discriminative features via contrastive activation: on corrupted input x^- (no grad) record $a^{\mathrm{corr}}_{\ell,f} = \max_t f_{\ell,t,f}$; on clean input x^+ (with grad) compute $a^{\mathrm{clean}}_{\ell,f}$ analogously. Task features satisfy

$$\Delta a_\ell \;=\; \mathrm{ReLU}\big(a^{\mathrm{clean}}_\ell - a^{\mathrm{corr}}_\ell\big) \;>\; \tau_\ell, \tag{1}$$

with τ_ℓ the per-example 95th percentile of positive Δa. The target MLP output contribution is $y^{\mathrm{target}}_\ell = D_\ell\big(f^{\mathrm{clean}}_\ell \odot \mathbf{1}_{\Delta a > \tau}\big)$, and the per-layer alignment loss is $\mathcal{L}_\ell = -\sum_t \langle h^{\mathrm{mlp,out}}_{\ell,t}, y^{\mathrm{target}}_{\ell,t}\rangle$. One backward pass over $\mathcal{L} = \sum_\ell \mathcal{L}_\ell$ accumulates $|\nabla_W \mathcal{L}|$ (sample_abs semantics, matching TaCQ). Only weights on paths that implement the selected task-discriminative features receive gradient; polysemantic CE would additionally reward weights tied to non-task features active at the same tokens.

Basis 1: transcoders (ATLAS-T). $F_\ell \gg d_{\mathrm{mlp}}$ sparse TopK transcoder features (facebook/crv-8b-instruct-transcoders; F=131,072), whose entries are approximately monosemantic [Bricken et al., 2023, Dunefsky et al., 2024]. This basis offers the cleanest feature separation but exists only for models with a published dictionary.

Basis 2: native MLP neurons (ATLAS-N, dictionary-free). We take the model's own MLP hidden layer as the feature basis: the post-activation intermediate $f_\ell = \sigma(W_{\mathrm{gate}}x) \odot (W_{\mathrm{up}}x) \in \mathbb{R}^{d_{\mathrm{mlp}}}$ serves as the code, and the model's own down_proj matrix is the decoder D_ℓ. The contrastive selection rule and alignment loss are identical to the transcoder case; only the basis changes.

Intuition. Individual MLP neurons are more polysemantic than dictionary features, so each selected neuron is a noisier task indicator. But the contrastive Δa filter operates at the population level: neurons that consistently fire more on clean than corrupted task inputs are, in aggregate, enriched for the task circuit even if no single neuron is monosemantic. Because the conjunction with CE saliency (below) demotes weights that only one signal favors, residual polysemantic noise in the circuit term is filtered rather than propagated into the mask. Empirically this native basis recovers most of the transcoder's gain on GSM8K and surpasses it on MMLU-STEM (§7.3)—the dictionary is an optional booster, not a requirement. ATLAS-N thus applies to any transformer with no external artifacts, removing the main practical barrier to task-circuit PTQ.

5.2 Conjunctive combination and mask

For each weight tensor W, define per-weight CE saliency g_{CE} as in TaCQ and circuit saliency g_{align} from $\mathcal{L}$ above (same $|W|$ and $|W - W_{b\mathrm{bit}}|$ scaling). Our main variant (mult) combines both via geometric mean:

$$S(W) \;=\; |W| \cdot |W - W_{b\mathrm{bit}}| \cdot \sqrt{g_{\mathrm{CE}}(W)\, g_{\mathrm{align}}(W)}, \tag{2}$$

selecting weights salient under both task loss and task circuit. The conjunction is the active ingredient: replacing g_{CE} entirely with g_{align} (align) or using CE alone ties or trails on every task we test, and union-style boosts ($g_{\mathrm{CE}}(1 + \lambda g_{\mathrm{align}})$) degenerate at 2-bit (§7.3). Global top-k over all eligible linear weights yields the binary mask M; GPTQ quantizes $M(W)$=0 to b-bit and keeps $M(W)$=1 in FP16.

Why the conjunction works at low bit-width. At 2-bit, quantization noise is large enough that protecting the wrong weights has a first-order cost: budget spent on weights that merely encode generic surface statistics (high CE saliency, low circuit saliency) or on circuit edges whose quantization error happens to be benign (high circuit, low CE) is budget taken from weights that are both functionally necessary and fragile. The geometric mean implements an AND-gate over the two failure modes. At 3-bit the noise floor drops and nearly any sensible task-conditioned mask suffices—which is exactly what we observe (§7.4).

5.3 Anchored union: consensus core and dispute region

Empirically, TaCQ and ATLAS-T agree on only ∼46–50% of the FP16 budget (J≈0.30–0.33; §9.1). We formalize a two-region policy that never renegotiates consensus outliers and allocates only the remaining budget on the method-disputed mass. Let Ω be the eligible weight indices, $s_{\text{TaCQ}}(w)$ and $s_{\text{ATLAS-T}}(w)$ the final per-weight scores, and $k = \lfloor \rho|\Omega| \rfloor$, ρ=0.0035. Each method induces a mask by global top-k; define $M_{\text{core}} \triangleq M_{\text{TaCQ}} \cap M_{\text{ATLAS-T}}$ and $M_{\text{dispute}} \triangleq M_{\text{TaCQ}} \triangle M_{\text{ATLAS-T}}$. Stage 1 locks the core ($M \supseteq M_{\text{core}}$); Stage 2 fills the residual budget $k_{\text{fill}} = k - |M_{\text{core}}|$ from M_{dispute} by ranking the blended normalized score $s_\lambda(w) = \lambda\, \hat{s}_{\text{ATLAS-T}}(w) + (1-\lambda)\, \hat{s}_{\text{TaCQ}}(w)$. The sweep over λ tests where the optimal dispute allocation lies as quantization severity changes (§7).

Implementation notes. Transcoders (when used) stream from CPU; saliency ranks on CPU. Native-basis extraction adds no memory overhead beyond hooks on the MLP intermediate. Model: Llama-3.1-8B-Instruct. Complexity matches TaCQ asymptotics (one forward–backward per calibration example), plus per-layer encode/decode for the transcoder basis only. [TODO: Report wall-clock vs. TaCQ on 128 pairs; add Algorithm 1 pseudocode.]

6 Experimental Setup

Model and interpretability stack. Llama-3.1-8B-Instruct via the unsloth mirror (bit-identical to Meta weights). Transcoders (for ATLAS-T only): facebook/crv-8b-instruct-transcoders (32 layers, TopK, F=131,072). ATLAS-N uses no external artifacts. All head-to-head comparisons use this model; Llama-3-8B TaCQ numbers reproduce the original paper on a separate track.

Tasks and metrics (TaCQ paper protocol).

- Spider text-to-SQL: execution accuracy on dev (1034 examples, test-suite-sql-eval).
- GSM8K: 8-shot chain-of-thought; flexible numeric extraction.
- MMLU: 5-shot MCQA; three subject splits (STEM, humanities, social sciences), each conditioned separately; calibration on the first 75% of each subject's test CSV, evaluation on the last 25%.

Calibration (task-conditioned). Corrupt model, importances, mask, and masked GPTQ are calibrated on the same task as evaluation. Spider uses the train split (1360 contrastive pairs); GSM8K uses train-split 8-shot CoT prompts (128 pairs); MMLU uses the test-0–75% rows (128 pairs). Evaluation rows are never used for calibration; an automated audit enforces the split policy on every run.

Multi-seed protocol. Seeds vary the contrastive calibration sample (and gradient/GPTQ RNG), not the quantizer or budget. Spider: four seeds for both methods. GSM8K/MMLU: three seeds (0–2).

Quantization protocol (matched for fairness). Outlier fraction 0.35% (mask_fraction=0.0035) over 224 attn+MLP weight tensors; global top-k; reference ΔW from b-bit RTN dequantization (ablation arms) or GPTQ corrupt checkpoint (Spider track); final quantization with GPTQ true-sequential, 128 calibration samples; bit-widths 2 and 3. Baselines: FP16, uniform GPTQ (no mask), TaCQ (contrastive CE saliency).

Hardware and reproducibility. All experiments run on 4×H200 (141 GB) nodes; replication scripts under scripts/ and TACQ/. [TODO: Report total GPU-hours.]

Table 2: Spider execution accuracy (%) on Llama-3.1-8B-Instruct, 0.35% budget, 1360 Spider-train contrastive calibration pairs. Multi-seed: mean ± sample std over four calibration seeds.

Method	FP16	2-bit
FP16 (unquantized)	67.8	–
TaCQ [Xiao et al., 2025] (mean)	–	26.5 ± 2.0
circuit-only (transcoder, no CE)	–	25.0
ATLAS-T (mean)	–	29.4 ± 2.7
ATLAS-T (best seed)	–	32.6

Table 3: Task-conditioned 2-bit (0.35% budget): mean over calibration seeds $\{0, 1, 2\}$. GSM8K: 8-shot CoT accuracy; MMLU: held-out last 25% of test.

Task	FP16	TaCQ	ATLAS-T	ATLAS-N
GSM8K	69.3	29.4	26.5	21.9*
MMLU STEM	59.0	44.5	43.6	41.9
MMLU humanities	66.0	51.1	44.7	[TODO: pending]
MMLU social sciences	75.6	57.2	52.8	53.6

*s2 pending (2 seeds). TaCQ = TaCQ firm; ATLAS-T = ATLAS-T; ATLAS-N = ATLAS-N (dict-free). Per-seed numbers in Appendix ??.

7 Results

We present results in the order of the paper's narrative: first, the transcoder-guided method (ATLAS-T) beats TaCQ at matched budget (§7.1, §7.2); second, a controlled ablation dissects why, revealing the conjunction as the active ingredient and the dictionary as optional (§7.3); third, the bit-width analysis locates where any of this matters (§7.4).

7.1 Spider text-to-SQL (primary head-to-head)

Table 2 reports execution accuracy on the Spider dev set (1034 examples, test-suite-sql-eval) under the matched protocol. ATLAS-T averages 29.4% over four paired calibration seeds vs. TaCQ 26.5% (+2.9 pp mean paired Δ); the best seed reaches 32.6%. Replacing CE entirely with transcoder alignment (the circuit-only arm) ties TaCQ (25.0 vs. 26.0)—a first hint, confirmed by the ablation below, that the improvement requires the conjunction, not the transcoder signal alone. At 3-bit, TaCQ reaches 64.8%, circuit-only ties (65.0%), and ATLAS-T reaches 66.2% (seed 0)—again a compressed gap at the milder bit-width.

Replication sanity check (Llama-3-8B). Our harness reproduces TaCQ Spider exec accuracy: 67.7 (FP16), 22.3 (2-bit), 60.1 (3-bit) vs. paper 67.6 / 21.92 / 58.32 [Xiao et al., 2025].

7.2 Task-conditioned downstream comparison (multi-seed)

Tables 3 and 4 report the firm head-to-head between TaCQ and ATLAS-T under the task-conditioned protocol (calibrate and evaluate on the same task; §6), with seeds varying the contrastive calibration sample.

7.3 Saliency-source ablation: what carries the gain?

Having established that ATLAS-T beats TaCQ at matched budget, we ask which ingredient does the work. The ablation holds everything fixed—model (Llama-3.1-8B-Instruct), FP16 budget (0.35%, global top-k), contrastive calibration pairs (128), GPTQ (true-sequential), and evaluation—and varies only the saliency signal that ranks weights. Eight arms span the hypothesis space from task-agnostic controls to the full conjunction, with each circuit arm instantiated in both the transcoder basis and the model's native MLP-neuron basis (Table 5).

Table 4: Task-conditioned 3-bit: same protocol. TaCQ and ATLAS-T over seeds $\{0, 1, 2\}$; ATLAS-N over seeds $\{0, 1, 2\}$. All gaps are within seed spread: at 3-bit the methods are statistically tied.

Task	FP16	TaCQ	ATLAS-T	ATLAS-N
GSM8K	69.3	63.7	61.4	57.9
MMLU STEM	59.0	56.5	54.6	53.6
MMLU humanities	66.0	63.0	[TODO: hfx]	[TODO: hfx]
MMLU social sciences	75.6	73.0	74.0	72.6

Table 5: Saliency-source ablation at 2-bit (0.35% FP16 budget, seed 0). Identical pipeline in every row; only the weight-ranking signal differs. GSM8K: 8-shot CoT accuracy. MMLU splits: 5-shot accuracy on the held-out last 25% of each subject's test split (chance = 25%). Dict. = requires an external feature dictionary.

Arm	Saliency signal	Dict.	GSM8K	STEM	Hum	Soc
random	uniform random	–	2.5	30.3	25.4	25.4
weight	$\lvert W\rvert$	–	14.4	33.6	33.9	38.7
magnitude	$\lvert W\rvert \cdot \lvert\Delta W_{\text{quant}}\rvert$	–	16.7	27.8	25.7	31.5
ce (≈TaCQ)	CE gradient saliency	–	25.4	41.3	44.2	54.1
align	circuit, transcoder basis	✓	25.6	39.2	42.2	49.0
$\text{align}_{\text{free}}$	circuit, native basis	–	22.9	42.6	45.7	52.6
mult (ATLAS-T)	CE ∩ circuit, transcoder	✓	28.7	38.9	45.0	53.4
$\text{mult}_{\text{free}}$ (ATLAS-N)	CE ∩ circuit, native	–	26.5	44.2	45.9	55.9

Finding 1: task-circuit information dominates bit-placement heuristics. Across all four tasks, every circuit-informed arm beats every control by a wide margin. On GSM8K, magnitude selection recovers only 14.4% and magnitude×error 16.7%, versus 22.9–28.7% for circuit-informed selection; random placement collapses to 2.5%. On MMLU the controls sit near chance (25–39%) while circuit arms reach 38.9–55.9%. Where the task circuit lives carries most of the recoverable signal at 2-bit.

Finding 2: the conjunction is the active ingredient. On GSM8K the conjunction beats both of its parts: mult 28.7% vs. ce 25.4% (loss-only) and align 25.6% (circuit-only); paired bootstrap p=0.005 for ATLAS-T vs. TaCQ. On MMLU-STEM the same holds in the native basis: $\text{mult}_{\text{free}}$ 44.2% vs. ce 41.3% and $\text{align}_{\text{free}}$ 42.6% (p=0.022 for ATLAS-N vs. ATLAS-T, subject-cluster bootstrap). On MMLU social sciences, ATLAS-N again leads (55.9 vs. ce 54.1 vs. $\text{align}_{\text{free}}$ 52.6). Neither signal alone suffices; the AND-gate over task loss and task circuit is what buys accuracy. This retroactively explains the Spider circuit-only result (§7.1): align-only tied TaCQ because it had discarded the loss half of the conjunction.

Finding 3: the dictionary is optional—and sometimes harmful. ATLAS-N recovers 92% of the transcoder conjunction on GSM8K (26.5 vs. 28.7) and surpasses every transcoder arm on MMLU-STEM (44.2 vs. 38.9 for mult). We attribute the reversal to basis–task fit: the CRV transcoders were trained to reconstruct general instruct-model behavior, and their feature space may tile math-word-problem structure (GSM8K) better than the heterogeneous knowledge retrieval MMLU requires; the native neuron basis is task-neutral. Practically, the external dictionary is an optional booster on some tasks rather than a dependency: conjunctive task-circuit PTQ works out-of-the-box on any transformer.

7.4 Conjunctive saliency is a low-bit phenomenon

We run the same controlled ablation at 3-bit (Table 6). The contrast with the 2-bit table is the paper's sharpest result.

At 2-bit (Table 5), the saliency source decides between collapse and function: the spread between conjunction and magnitude controls is 10–26 points, and weight magnitude recovers only 14.4% on GSM8K. At 3-bit, these gaps collapse: weight magnitude leads on GSM8K (62.4% vs. mult 60.9%), and all circuit

Table 6: Saliency-source ablation at 3-bit (0.35% FP16 budget, seed 0). Same eight-arm protocol as Table 5; only the bit-width changes.

Arm	Saliency signal	GSM8K	STEM	Hum	Soc
random	uniform random	36.2	45.4	52.3	60.1
weight	$\lvert W\rvert$	62.4	54.2	58.5	70.1
magnitude	$\lvert W\rvert \cdot \lvert\Delta W\rvert$	56.6	48.7	56.4	65.9
ce (≈TaCQ)	CE gradient	58.3	54.7	60.7	71.2
align	circuit, transcoder	59.0	54.5	62.0	72.1
mult (ATLAS-T)	CE ∩ circuit, transcoder	60.9	54.2	62.2	74.2

Table 7: 2-bit Spider dev exec under native gptq.llama (128 samples, true-sequential). Same quantizer; only mask identity and calibration corpus vary.

Outlier mask	GPTQ calibration	Exec (%)
TaCQ MSG	C4	0.6
TaCQ MSG	Spider (128)	26.0
B0 heavy (ATLAS-T)	C4	0.0
TaCQ full pipeline (Spider contrastive 1360)		26.0

arms cluster within 4 points on every task. The mechanism is simple: 3-bit quantization noise is small enough that GPTQ's column-wise error compensation absorbs most task damage regardless of which 0.35% of weights are protected. At 2-bit the noise is catastrophic and the FP16 budget must land on weights that are simultaneously task-critical and quantization-fragile—precisely what the conjunction selects.

The quantizer–saliency interaction. To verify that the gain is in the mask, not the quantizer, we replace GPTQ with RTN (round-to-nearest: no calibration, no error compensation) while keeping the same masks. At 2-bit RTN, every mask—including our best—produces 0% GSM8K accuracy and 22% MMLU-STEM (uniform random chance). RTN's rounding error at 2-bit is catastrophic and unredeemable: no outlier selection can help because the 99.65% of quantized weights are too noisy for the model to function. This locates the saliency contribution precisely: it operates in the window that GPTQ opens—between "noise too mild for masks to matter" (3-bit) and "noise too severe for anything to help" (2-bit RTN).

A cautionary note on bit-width bookkeeping. During this study we found that an earlier run had silently built "2-bit" ATLAS-T engines at 3 bits (a quantizer default overriding the pipeline intent), producing implausibly strong 2-bit numbers (∼62% GSM8K) that a matched-budget ablation immediately exposed (28.7%). We recommend controlled saliency-source ablations as standard practice for mixed-precision PTQ papers: they are cheap, and they catch both bookkeeping errors and overclaiming.

Dispute sweep (anchored union). The anchored-union construction (§5.3) locks the consensus core (∼46% of budget at 2-bit) and re-ranks only the XOR dispute mass. The mask morph behaves as designed (J vs. TaCQ falls from 0.708 to 0.462 as λ goes 0 to 1). Partial Spider evals at 3-bit peak at the TaCQ-heavy endpoint (λ=0: 63.2% vs. 59.9% at λ=0.5), consistent with milder quantization favoring load-bearing MLP mass over routing-heavy circuit weights. [TODO: complete 2-bit sweep.]

8 Domain-shift at 2-bit: calibration corpus controls

Standard GPTQ pipelines often calibrate on generic web text (e.g., C4). We isolate the effect of calibration domain at fixed 2-bit width, mask budget (0.35%), and gptq.llama implementation, using pre-computed masks without re-running saliency.

Table 7 shows a 25+ point collapse when switching MSG from Spider to C4 calibration at 2-bit, while the closing Spider-native control reproduces the full TaCQ 26.0% baseline. Generic corpus calibration is therefore a confound in 2-bit task evaluations; task-matched calibration is necessary even when the outlier mask is held fixed.

9 Analysis

9.1 Mask overlap: a different circuit, not a reranking

Both ATLAS-T and TaCQ keep exactly 0.35% of the 224 attn+MLP weight tensors—24.4M FP16 outliers each—yet they select largely disjoint subsets. Against TaCQ 2-bit (seed-matched), only 46% of ATLAS-T outliers overlap (J=0.298); at 3-bit the overlap is 50% (J=0.331). Thus ∼54% of ATLAS-T's FP16 budget protects weights TaCQ would quantize—this is not a minor reranking within the same circuit (Figures 1–2).

Module-level divergence. MLP blocks contribute the largest TaCQ-exclusive mass (∼2.9–3.8M per projection), while ATLAS-T retains more attention weights (q/k/v_proj exceed TaCQ-only by ∼0.4–1.3M per head type). Aggregated, ATLAS-T adds 5.5M routing-centric outliers vs. 3.0M for TaCQ; TaCQ adds 10.2M MLP-only vs. 7.7M. The circuit term steers the budget toward mid-network routing while reshuffling which MLP channels stay in FP16.

Layer-wise divergence. Disagreement peaks in mid layers 4–14 (layer 9 most divergent); early and late layers agree more. The mid-stack is where task-specific schema and compositional structure are processed in Llama-family models.

Seed sensitivity vs. method divergence. Cross-seed Jaccard among ATLAS-T masks is 0.995–0.997 (only ∼36–56k of 24.4M kept weights differ per seed pair), vs. 0.983–0.985 for TaCQ (∼187–214k). Cross-method J=0.298. Methods select different circuits, not different seeds of the same circuit; TaCQ's polysemantic CE gradients amplify calibration-shuffle effects more than the conjunction.

9.2 Causal validation via targeted suppression

Scaling the 24.4M ATLAS-T-masked weights by α=0.8 on the FP16 model (no quantization) drops Spider exec from 67.8% to 38.9% while short non-SQL English probes remain fluent; the boost control (α=1.2) also hurts (−8.2 pt). The task-selective failure mode rules out the null that the mask tags generic high-magnitude weights. A leak-free contrastive squeeze (α_{task}=1.05, α_{base}=0.98, selected on a train holdout) yields +1.0 pt dev exec over contemporaneous FP16.

9.3 Why the native basis can beat the dictionary

The MMLU-STEM reversal (mult$_{\text{free}}$ 44.2 vs. mult 38.9; Table 5) suggests dictionary quality is task-dependent. Two non-exclusive explanations: (i) coverage — CRV transcoders are trained to reconstruct instruct-model MLP behavior on broad data; features that tile mathematical CoT structure (GSM8K) may be better resolved than the fine-grained factual retrieval MMLU stresses, so the transcoder's Δa filter passes a noisier task set on MMLU; (ii) reconstruction error — the alignment loss targets the transcoder's approximation of the MLP output; where reconstruction is weak, gradients partially chase approximation artifacts. The native basis has zero reconstruction error by construction (its decoder is down_proj), trading monosemanticity for faithfulness. The conjunction with CE then filters its polysemantic noise. This trade-off predicts the dictionary helps where its training distribution matches the task and the native basis is safer elsewhere—a testable criterion for future dictionary-aware PTQ.

9.4 Efficiency and failure modes

Saliency extraction matches TaCQ asymptotics; the transcoder basis adds per-layer encode/decode, the native basis adds only forward hooks. Union-style combinations ($g_{\text{CE}}(1+\lambda g_{\text{align}})$) degenerate at 2-bit (model

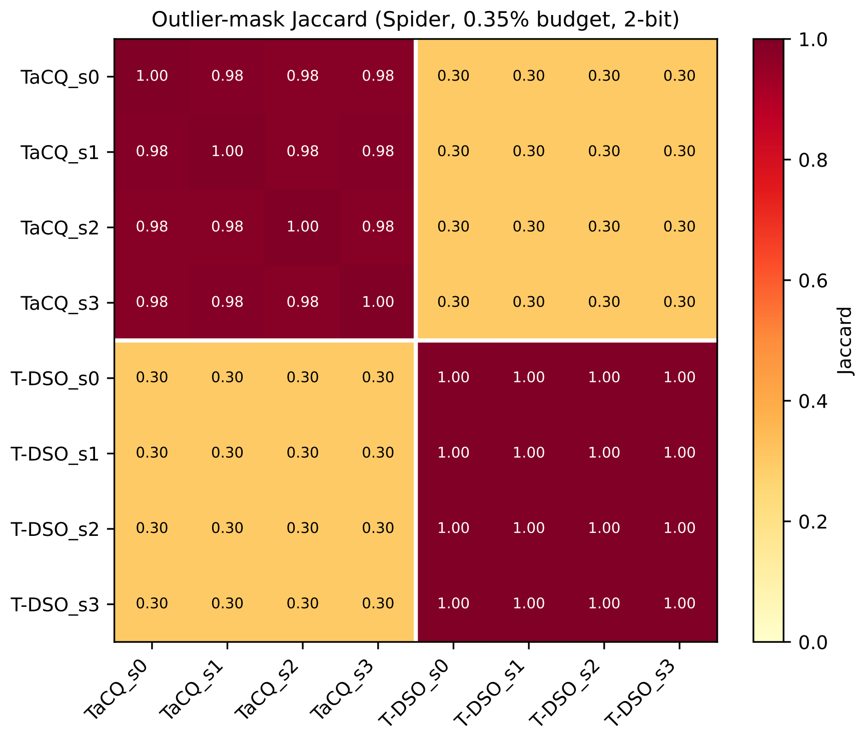

Figure 1: Cross-seed and cross-method Jaccard among 0.35% outlier masks (Spider, 2-bit). Diagonal blocks: within-method seeds ($J > 0.98$); off-diagonal: method gap ($J \approx 0.30$).

rambles, no EOS), confirming that expanding the CE mask with circuit bonuses is worse than intersecting the signals.

10 Limitations

- Dictionary variant coverage. Full per-layer CRV transcoders exist only for Llama-3.1-8B. The dictionary-free variant (ATLAS-N) removes this dependency for the method itself, but our cross-basis comparison (transcoder vs. native) is correspondingly limited to this model; replicating the basis–task interaction on a second model family is in progress.
- Approximate MLP targets (transcoder basis). TopK transcoders reconstruct MLP output imperfectly; where reconstruction is weak the alignment gradient partially chases approximation artifacts (§9.3). The native basis avoids this by construction.
- Seed / calibration variance. Both methods vary with calibration seed at 2-bit (Spider: ATLAS-T spans 26.2–32.6%, TaCQ 25.0–29.6%). We report paired multi-seed means; single-seed entries are marked. Paired bootstrap confidence intervals are pending.
- Task coverage. Firm evidence on Spider, GSM8K, and three MMLU splits; HumanEval and broader reasoning suites are future work.
- Scale. Results are on an 8B model. The dictionary-free variant makes scale extension independent of dictionary releases; 70B validation is queued.

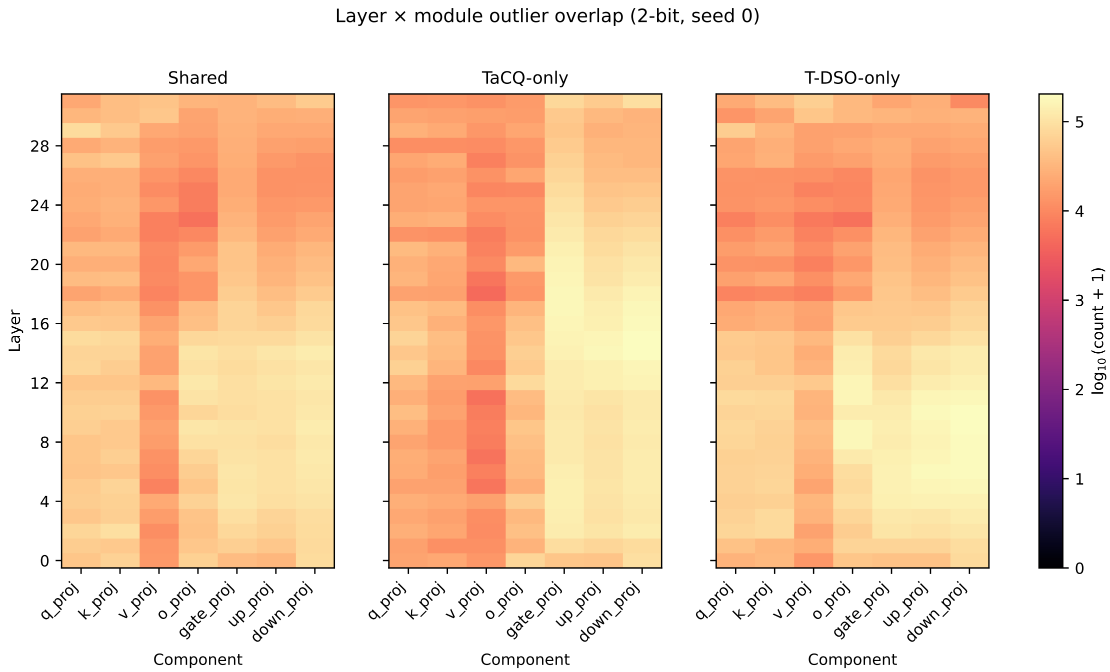

Figure 2: Layer × module outlier decomposition (2-bit, seed 0): shared, TaCQ-only, and ATLAS-T-only kept weights.

11 Systems Outlook: Dynamic Sparse Overlays

Beyond static PTQ comparison, ATLAS-T motivates a Dynamic Sparse Overlay execution model not available to TaCQ. We do not claim these systems results in this paper; they define the research arc if the dissociation holds empirically.

Agentic hot-swapping. TaCQ produces one task-conditioned mask baked into the quantized checkpoint—configuration is fixed for the inference job. If sub-skills (schema linking, SQL generation, self-correction) prefer different outlier sets, TaCQ must compromise with a single global top-k. ATLAS-T's feature-space objective naturally supports separate calibration streams per sub-skill and, with an orthogonality penalty on mask overlap, non-redundant overlays M_{link}, M_{gen}, M_{fix}. At runtime, only the FP16 outlier values differ; the 2-bit base $W_{2\mathrm{bit}}$ stays resident while overlay tables stream (e.g., via PCIe into on-chip SRAM). [TODO: Formalize QAL + orthogonality penalty; implement mask streaming benchmark.]

KV-cache inheritance. Multi-step text-to-SQL pipelines often chain agents as separate API calls. Each call re-tokenizes history and recomputes attention from scratch—$O(N^2)$ per step. With a shared frozen $W_{2\mathrm{bit}}$ and hot-swapped overlays, a downstream sub-circuit inherits the prior step's KV cache; only overlay weights and the active ATLAS-T mask change. Time-to-first-token for steps 2+ approaches cache-append cost rather than full prefill. [TODO: Microbenchmark TTFT: static TaCQ vs. overlay swap on two-step Spider prompt.]

Positioning for reviewers. Dimensions 2–3 are enabled by feature dissociation plus mixed-precision structure; this paper validates Dimension 1 (Spider exec accuracy at matched budget). Systems claims require separate engineering evidence and belong in an extended version or companion systems section once built.

12 Conclusion

Low-bit PTQ forces a choice about which tiny fraction of weights must remain in full precision. We started from an interpretability-guided answer—route saliency through sparse transcoder features and intersect with the task loss (ATLAS-T)—which beats TaCQ at matched budget. Dissecting it with a controlled ablation revealed the general principle underneath: protect a weight only if it is salient to the task loss and to the activation circuit separating clean from corrupted task inputs, where the circuit can be defined in the model's own feature basis (ATLAS-N).

Three takeaways. (1) The conjunction is the active ingredient. In a controlled saliency-source ablation with everything matched except the ranking signal, the conjunction beats loss-only (CE/TaCQ) and circuit-only selection on both GSM8K and MMLU-STEM at 2-bit, and all circuit-informed arms dominate magnitude/error/random controls by 10–26 points. (2) No dictionary required. Computing the circuit term in the model's own MLP-neuron basis (ATLAS-N) recovers most of the transcoder variant's gain on GSM8K and surpasses it on MMLU-STEM—turning task-circuit PTQ from a method gated on interpretability artifact releases into one that applies to any transformer out-of-the-box. (3) Low-bit is where saliency matters. At 3-bit all task-conditioned methods converge to within seed noise; at 2-bit the saliency source separates collapse from function. Mixed-precision PTQ research should therefore be evaluated—and ablated—at the bit-widths where selection actually binds.

On Spider, intersecting CE with transcoder task-circuit attribution selects a markedly different circuit (J=0.30 vs. TaCQ) and improves 2-bit execution accuracy by +2.9 points on average at matched budget, with causal knockout supporting that the selected subnetwork implements task routing.

Future work. Multi-task overlay masks (§11), paired-bootstrap significance on all headline gaps, second-model replication of the basis–task interaction, and budget-fraction sweeps locating where the conjunction's advantage emerges.

Reproducibility. Code, masks, contrastive calibration sets, and replication logs: [TODO: anonymous repo URL for submission].

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A Hyperparameters

Table 8: Shared hyperparameters across TaCQ, ATLAS-T, and ATLAS-N.

Parameter	Value
Outlier fraction	0.35% (global top-k)
Calibration pairs	128 (Spider track: 1360)
Max sequence length	2048
Feature threshold quantile	0.95 (per-example, positive Δa)
Transcoder TopK	128 (ATLAS-T only)
Native feature basis	MLP intermediate, d_{mlp}=14,336 (ATLAS-N)
GPTQ	true-sequential, 128 samples
ΔW reference	RTN b-bit (ablation) / GPTQ corrupt (Spider)

B Contrastive corruption schema

Spider: schema swaps, wrong-SQL completions. GSM8K: corrupted chain-of-thought with wrong intermediate steps and final answer. MMLU: wrong answer-choice completions under the same 5-shot prompt. [TODO: full schema table from data_prep_contrastive.py]

C Saliency-source ablation: arm definitions

All arms share the mask format, 0.35% global top-k budget, GPTQ, and evaluation. random: uniform random scores. weight: $|W|$. magnitude: $|W| \cdot |\Delta W_{\text{RTN-}b\text{bit}}|$. ce: $|W| \cdot |\Delta W| \cdot g_{\text{CE}}$. align / align$_{\text{free}}$: $|W| \cdot |\Delta W| \cdot g_{\text{align}}$ in the transcoder / native basis. mult / mult$_{\text{free}}$: Eq. 2 in the transcoder / native basis.

D Transcoder-native feature audit

Per-layer task-feature density (meta.layer_feature_density) peaks in mid layers 4–14, consistent with the layer-wise weight disagreement (§9.1). Top-K feature ranking by cumulative clean−corrupt activation Δa and manual labeling templates are shipped with the code. [TODO: populate manual labels]

E Additional results

[TODO: Full Spider difficulty breakdown; complete dispute-sweep 2-bit row; RTN vs. GPTQ ΔW ablation; cross-seed Jaccard tables.]

F Broader impact

Compression lowers the cost of deploying capable LLMs, which is dual-use in the usual ways. Our interpretability-guided selection does not introduce new capability risks beyond standard PTQ; it may improve auditability by making the protected task circuit explicit.